

USE CASE FLOW

Goal	Process Sale Transaction
Actor	Staff
Precondition	Staff is logged in, and the selected product exists in the inventory with sufficient stock.
Main Flow	1. Staff selects New Sale from the dashboard. 2. The system displays the product search screen. 3. Staff searches for and selects one or more products to add to the cart. 4. The system validates the stock availability for each selected item. 5. Staff confirms the required quantity for each item. 6. The system calculates the subtotal, tax, and final total. 7. Staff clicks Checkout. 8. The system deducts the sold quantity from the inventory stock. 9. The system generates an invoice containing the transaction ID, date, itemized product list, and total amount. 10. The system displays the invoice and provides an option to print it. 11. The sale transaction is completed.
Alternate Flow	Insufficient Stock At Step 4, if the requested quantity exceeds the available stock: 4a. The system displays an Insufficient Stock error message. 4b. The system prevents the item from being added to the cart with the requested quantity. 4c. Staff must correct the quantity before continuing with the transaction. Then move to step 5. Low Stock Alert At Step 11, after the sale is completed: 11a. The system checks the remaining stock level. 11b. If the resulting stock falls below the configured reorder threshold, the system triggers a low-stock alert. 11c. The Manager is notified about the low-stock product. End
Postcondition	The inventory is updated, an invoice is generated, and an audit log entry is created.

Goal	Manage Products
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Actor	Admin
Precondition	Admin is logged in.
Main Flow	1. Admin navigates to the Product Management screen. 2. The system displays the list of existing products. 3. Admin chooses to add, edit, or delete a product. Add Product 1. Admin enters the product details, including name, SKU, category, price, quantity, and supplier. 2. The system validates that the SKU is unique. 3. The system saves the new product record. Edit Product 1. Admin selects an existing product. 2. Admin modifies the required product details. 3. The system validates and updates the product record. Delete Product 1. Admin selects a product for deletion. 2. Admin confirms the deletion. 3. The system removes the product record. 4. The system records the performed action in the audit trail. 5. The system refreshes the product list.
Alternate Flow	Duplicate SKU At Step 3 (Add Product), if the entered SKU already exists: 3a. Admin attempts to add a product using an existing SKU. 3b. The system detects the duplicate SKU. 3c. The system displays an appropriate error message. 3d. The product creation process is cancelled until a unique SKU is provided. End of Add Product
Postcondition	The product catalog is updated, and an audit log entry is created.

Goal	Manage Users and Roles
Actor	Admin
Precondition	Admin is logged in.
Main Flow	1. Admin navigates to the User Management module.

	2. Admin creates a new user account by entering a username, password, and role. 3. The available roles include Admin, Manager, and Staff. 4. The system validates that the username is unique. 5. The system assigns the appropriate role-based permissions. 6. Admin can modify an existing user's role. 7. Admin can deactivate or remove a user account. 8. The system updates the user's access permissions immediately.
Alternate Flow	Username Already Exists At Step 4, if the entered username already exists: 4a. Admin enters a username that already exists in the system. 4b. The system rejects the new account creation. 4c. The system displays an appropriate error message. Then move to step 5.
Postcondition	The user account is created or updated with the correct role-based access permissions.

Goal	Manage Suppliers
Actor	Admin
Precondition	Admin is logged in.
Main Flow	1. Admin navigates to the Supplier Management module. 2. Admin enters the supplier details, including the supplier name, contact information, and address. 3. Admin links the supplier to one or more relevant products. 4. The system saves the supplier record. 5. The system updates the references between the supplier and the linked products.
Postcondition	The supplier catalog is updated, and the supplier is linked to the relevant products.

Goal	Create Purchase Order
Actor	Manager

Precondition	Manager is logged in, and the required product and supplier records exist.
Main Flow	1. Manager views products with low stock or manually selects products. 2. Manager selects the required product or products. 3. Manager specifies the required quantity for each product. 4. Manager selects the appropriate supplier. 5. The system generates a Purchase Order with the status set to Pending. 6. Manager reviews the Purchase Order details. 7. Manager confirms the Purchase Order. 8. The system saves the Purchase Order. 9. The system notifies the relevant stakeholders.
Postcondition	A new Purchase Order is created with the status Pending.

Goal	Receive Purchase Order
Actor	Manager
Precondition	A Purchase Order with the status Pending exists.
Main Flow	1. Manager opens the Purchase Orders list. 2. Manager selects the relevant Purchase Order. 3. Manager marks the Purchase Order as Received. 4. The system validates the received quantity against the ordered quantity. 5. The system updates the product stock according to the received quantity. 6. The system changes the Purchase Order status to Received. 7. The system records the transaction in the system log.
Alternate Flow	Partial Delivery At Step 4, if the supplier delivers only part of the ordered quantity: 4a. The supplier delivers only part of the ordered quantity. 4b. Manager enters the actual quantity received. 4c. The system validates the partial delivery details. 4d. The system increases the product stock according to the quantity received. 4e. The system changes the Purchase Order status to Partially Received. 4f. The remaining quantity continues to be pending. Then move to step 5.

Postcondition	The inventory stock is increased according to the received quantity, and the Purchase Order status is updated.
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Goal	Generate Reports
Actor	Manager
Precondition	Manager is logged in, and relevant sales and stock data exists in the system.
Main Flow	1. Manager navigates to the Reports module. 2. Manager selects the required report type. 3. The available report types include Sales, Stock, and Revenue reports. 4. Manager selects the required date range. 5. The system collects and processes the relevant data. 6. The system calculates and aggregates the required information. 7. The system generates and displays the report. 8. Manager can export the report in CSV or PDF format.
Postcondition	The requested report is generated successfully and is available for viewing or download.

Goal	View Low Stock Alerts
Actor	Manager
Precondition	Reorder thresholds are configured for the products.
Main Flow	1. The system monitors product stock levels after each inventory transaction. 2. When a product's stock falls below its configured reorder threshold, the system identifies the product as low in stock. 3. The system displays a low-stock alert on the dashboard or notification panel. 4. Manager views the low-stock alert. 5. Manager can initiate the creation of a Purchase Order directly from the alert.
Postcondition	The Manager is notified about products with low stock and can take immediate action to replenish them.

Goal	Search and Filter Products
Actor	Staff
Precondition	Staff is logged in, and products exist in the system.
Main Flow	1. Staff enters a search term or applies one or more filters. 2. Staff can search or filter products based on category, stock status, or SKU. 3. The system queries the product database using the provided search criteria. 4. The system retrieves the matching products. 5. The system displays the matching results to the Staff member.
Alternate Flow	No Products Found At Step 4, if no products match the search criteria: 4a. No products match the entered search term or selected filters. 4b. The system displays the message No products found. Then move to step 5.
Postcondition	Relevant products matching the search criteria are displayed to the Staff member.

Goal	Login
Actor	All Users, including Admin, Manager, and Staff
Precondition	The user has a valid account in the system.
Main Flow	1. User enters their username and password. 2. The system validates the entered credentials against the database. 3. The system verifies the user's assigned role. 4. The system establishes a valid user session. 5. The system redirects the user to the appropriate role-based dashboard.
Alternate Flow	Invalid Credentials At Step 2, if the entered credentials are invalid: 2a. User enters an incorrect username or password. 2b. The system detects that the credentials are invalid. 2c. The system displays an appropriate error message. 2d. The system denies access to the application. Then move to step 3.

Postcondition	The user is successfully authenticated and granted access according to their assigned role and permissions.
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